A review of practices suitable for ethics-aware software engineering

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Abstract

Research shows that the lack of ethical practices for regulating human behaviour and ethical judgment in software development is a concern. The lack of ethical practices leads to the development of software which is not in line with the ethical needs of software stakeholders. The consideration of the ethical needs of stakeholders and their integration into the software process requires knowledge of the relevant practices. This study sought to identify and synthesise ethical practices suitable for supporting ethics-aware software engineering. To achieve the objective of this study literature review was conducted based on search keywords. This study determined that ethics awareness, enforcement, decision-making, evaluation, and governance are integral ethical practices for achieving ethics-aware software engineering practice to cater for stakeholder ethical needs. Although the findings of this study are based on literature sources, future empirical studies should be conducted to validate the combination of these ethical practices to determine how they can support ethics-aware software engineering practice.

CCS Concepts

• **ethics-aware software engineering**; • **software development**; • **computing ethics**;

Keywords

ethics-aware software engineering, practices, framework

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1 Introduction

At the height of unprecedented software demand and usage, integrating stakeholders' ethical needs into the business goals and activities of organizations involved in software development is important. In addition, software users require transparency and accountability, as they are inclined to buy from businesses that embrace ethical practices [1]. Integrating ethical needs requires much exercise of ethical judgment and action [2]. Therefore, the allowance to specify and achieve the ethical requirements should be provided over and above the software meeting the specified functional requirements. To succeed in this endeavor, software environments must ensure ethical practices are prevalent in any aspect relating to software development. Ethical practices are good for regulating human behavior and advocating for the welfare of society and individuals affected by the software product. However, software engineering research shows that the integration of ethical practices into efforts of developing software remains a challenge.

The challenge of lack of ethical practices in the development of software has surfaced as a concern in both government and private sector organizations. This has led to the development and deployment of software that is lacking ethical alignment [3]. For example, a lack of ethical practices in software design resulted in failures that crashed Boeing Max 737 aircraft. In the analysis of the Boeing MAX aircraft case, Travis 2019 [4], a pilot and software engineer indicates that safety imperatives were overlooked and priority was given to business profits. In addition, a subsequent analysis of the case by Herkert et al. [5] determined that the engineers of the aircraft failed to consider the ethical impact of both the design decisions of their hardware and software. Consequently, other business needs such as time to reach the market prevailed over ethical practices. In terms of the government sector example, a health provision service resulted in unequal health care to patients due to the algorithm bias of a software used for determining health services to be provided to patients [6].

Although ethics are organizational practices, in software engineering the ethical practice is also the professional responsibility of the individual software practitioner. Therefore, the software environments and individuals involved in creating software are responsible for ensuring that ethical practices are followed. Ethical practices involve the integration of stakeholder ethical values in all dimensions of the software engineering practice for the outcomes of software development to reflect ethicality. Borrowing from the concept of ethical business practices advanced by [7], the practices should amongst others include ethical principles, standards, and expectations on the ethical conduct of all individuals in the organization.

Research scholars agree that for the software engineering practice to thrive, ethical practices should be at the forefront of all software development efforts. Furthermore, frameworks for ethics-aware software engineering require the inclusion of ethical practices to improve ethical software development [8], [9]. Although

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ethical practices are the cornerstone of ethical software development, to the knowledge of the authors of this article, a review study aimed at identifying practices to inform the development of a framework for ethics-aware software engineering practice has not been done. The research question formulated to be answered by this study is: *What are the existing ethical practices suitable for supporting ethics-aware software engineering practice?*

To achieve the aim of this study, we used a literature review to identify studies from which we can extract aspects of ethical practices related to software engineering. Keywords were generated from the research question constructs. A combination of keywords was used to search databases such as Google Scholar, Scopus, and IEEE to identify relevant research articles.

The next section of the paper briefly describes some of the concepts related to ethics in software engineering. This is followed by a detailed account of the various ethical practices found in software engineering. The paper draws some implications and conclusions on the ethical practices as useful components of an ethics-aware software engineering practice.

2 Ethics and ethical practices in software engineering

2.1 Software engineering ethics

Ethics are generally concerned with the determination of what is good or bad about moral duty and obligation [10]. This is commonly based on principles agreed upon by a community of practice. In the computing field, ethics principles are encapsulated in codes of ethics published by professional bodies responsible for regulating the practice. The aim of the codes of ethics is to ensure that the ethical behavior of individuals and organizational environments responsible for software development is guided accordingly. It is so because users of software have trust in those creating software because of their specialized knowledge and skills. Gotterbarn [11] defines software engineering ethics as an interaction between technology (that is technical decisions to create such technology) and ethical principles used to advance human values.

According to Gotterbarn [11] an ideal software engineering ethics model should encompass ethical guidelines organized in ethical codes, software stakeholders' interactions with the software process, and areas within which ethical challenges arise in the software engineering practice. According to Harris, et al. [12] and Marcu et al. [13] there are different categories of ethics, such as professional, personal, and social/general ethics. In the expanded taxonomy of ethics applicable to software engineering there are general, professional, and technical ethics; and professional ethics outweigh the rest [11]. Although individuals involved in software development may have sourced their ethical knowledge and skills from general and technical ethics, professional ethics are more useful in organizational environments.

2.2 Ethical practices

Ethical practices are summed up as the integration of ethical values across all organizational operations. In terms of software development, this may include embracing ethics imperatives as part of the software project approvals, the application of ethical principles in the daily operations of software development organizational environments, and collaboration with stakeholders. This may also include making software teams understand the ethical expectations of the organization and that of the other software stakeholders.

The emergence of frontier technologies such as artificial intelligence, machine learning, and the Internet of Things (IoT) which enables the computational performance of tasks that are commonly performed by human beings, dictates the need for a concerted ethical approach to software development. This is mainly due to the ethical risks that may be brought by these technologies. Ethical practices are useful in regulating the conduct of stakeholders and ensuring that software development endeavors are aligned to ethical standards to allow for the ethical development of software. Therefore, a practice in software development environments to support ethics in the same way as the technical aspects of software are supported in developing software is important.

Based on the literature search and analysis, we have identified ethical practices which are summarized in Table 1 and discussed below. The identified practices are awareness, enforcement, decision-making, evaluation, and governance. We further show in Figure 1 the mapping of these ethical practices to the software process. In line with Gogoll et al. [14]'s assertion that ethics should be located deep into the software development process, this organisation demonstrates how ethics practices link with software engineering practices.

2.2.1 Ethics awareness. Ethics awareness denotes the extent to which a person can demonstrate knowledge of ethics codes and skills to apply them [41]. Software stakeholders must be aware of what is acceptable and unacceptable in terms of ethical principles applicable to the development of software. Therefore, it is incumbent on organizations to ensure that there is awareness of what is ethically acceptable or unacceptable. Research studies (e.g. [16], [30], [42]) concluded that for ethics to make a difference, software development environment awareness or organizational orientation to ethics is important. Organizations and individual practitioners need to put in place initiatives for ensuring ethical awareness. The initiatives should include the publication of policies and codes of ethics and their communication to the software development teams [8], [16]. Ethics codes are useful for reminding stakeholders about their responsibilities, thus promoting ethical behavior [19].

Furthermore, ethics awareness can be included in the employee onboarding process and can be part of the psychological contract in which the employer communicates ethical requirements or expectations [23]. In addition, an ethics culture can be systematically introduced in an organization to nurture learning which promotes ethical behavior [23]. Ethical leadership can also serve as a means for ethics awareness in organizations [10], [20–22]. These include fairness in decision-making, moral traits, and behavior which amounts to good role modeling [22]. This is important because management sometimes does not consider themselves to be bound by the codes of ethics [8]. All these awareness activities can be broadly taken as ethics institutionalization [23], [43], [44]. The awareness should be completed with how important or useful the stakeholders perceive the ethics to be in their software development work. For example, Wotruba [45] states stakeholders are inclined to consider codes of ethics useful when they understand them and their purpose. On that note, the study determined that familiarity

Table 1: Software engineering ethical practices

Ethical Practice	Literature sources	Constituent elements of the practice
Awareness	[8], [15–22, 22]	Communication, policy training, awareness initiatives, top management public statements about ethics, management leading by example, Institutionalization, organizational commitment, employee onboarding processes
Enforcement	[22–27]	Reporting of ethical violations, rewarding of ethical behavior, contributions, and loyalty, implementation of sanctions, communication of disciplinary sanction outcomes, ethics governance structures, punish unethical behavior.
Decision-making	[17], [18], [21], [24], [28–30]	Group decision-making, mechanisms such as ethical deliberation and ethical cycle, consultation with seniors for advice, and use of tools for decision-making.
Evaluation	[9], [27], [28], [31–37]	Assessment plans, ethical risk assessment, ethical impact assessment, ethical theories and principles, stakeholder needs, evaluation checklists, and formative or summative forms of assessment.
Governance	[7], [32], [38–40]	Ethics committees, leadership commitment, ethics officer, standardized reporting, accountability, laws and regulations, frameworks, tools, models and policies, monitoring, strategic planning, accountability contract, public engagement frameworks,

with codes enhances the perceived usefulness and adoption of the codes. Therefore, applying suitable awareness practices can enable organizations to achieve ethical objectives [18], [42], [46].

2.2.2 Ethics Enforcement. The enforcement of ethics is considered a useful ethical practice in achieving organizational objectives [24], [47]. The practice of ethics enforcement includes placing ethics entry requirements in terms of qualifications, skills, experience, and certifications that can enable potential employees to handle ethical challenges as they occur [23]. In addition, organizations should ensure that professional codes of ethics and organizational codes are made mandatory reference material and are accessible to guide ethical behavior and decision-making [24].

Enforcement can also be achieved through the establishment of platforms for reporting ethics violations or whistleblowing hotlines to support stakeholders when facing ethical challenges, recognition of employees' contributions through incentives, punishing unethical tendencies, and communication of practical disciplinary cases outcomes of unethical behaviors [22], [23], [25], [26] amongst others. Furthermore, the establishment of corporate ethics governance structures can contribute to the achievement of ethics enforcement to reach organizational goals [23]. Lastly, [27] posits that enforceable standards that are actionable and relatable should be developed. The enforcement of ethics supports organizational environments to put measures in place to police or monitor ethical activities.

2.2.3 Ethics decision-making. Ethical decision-making requires that when a person is confronted with a situation encroaching on ethics, the person should be able to apply their ethical knowledge and skills to conclude a decision. Furthermore, the ethical disposition or attitude of an individual is important in determining how far they are willing to apply their ethical knowledge and skills.

O'Fallon and Butterfield [48] and Pierce and Henry [30] submit that ethical decision-making should be supported through relevant sources to guide decisions. In software development, the profession has abounded with various codes of ethics to which members and the public can always refer to. Furthermore, organizations have internal codes of ethics which can be used to guide decision-making. In addition, those participating can also consult their senior management for advice to enrich their decision-making [21]. The practice also includes efforts to harmonize the different types of ethical sources such as personal, and professional codes with organizational codes [17], [24], [30].

To support ethical decision-making organizations, need to provide devices or mechanisms that are part of the practice. Example mechanisms such as the use of the use of ethical cycle which offers structure to inquiry and critical engagement is suggested by Morrison and Wallace [28]. The ethical cycle can be used in any software methodology. The other mechanisms of the practice are ethical deliberations [29], [49–54]. The ethical deliberations mechanism requires its users to have in-depth knowledge of the normative criteria used for considering the ethicality of software artifacts. It allows for answering difficult questions about the ethical principles being infringed and the stakeholders who are affected by the inclusion or lack of inclusion of such principles in the final software product. Therefore, ethical deliberations empower those involved in software development to independently and collectively engage in decision-making in a rational and organized manner.

Figure 1: Ethical practices in software engineering

2.2.4 Ethics Evaluation. Ethical considerations should form an important part of evaluating computerized systems [31]. Organizations should strive to allow the evaluation of ethics at different levels of the software process. Gotterbarn [11] suggests that a practice consisting of various tools be adopted to assist with ethics evaluation. The evaluations and the criteria used should factor in the interests and values of stakeholders [31]. By doing so, it will ensure that the ethical needs of various stakeholders are accounted for. The evaluation practice should be inspired by suitable ethical theories [28]. Amongst others, the practice should include ethical risk analysis[11], ethics impact assessment [9], [27], [32], stakeholder web of needs [33], and checklists [34], [35]. Furthermore, the evaluation can be conducted on a formative or summative basis. The summative evaluation is done at the end whilst the formative evaluation is done collaboratively to inform further developments in the system [31].

The evaluation will provide certainty to those affected by software that their needs have been considered. Secondly, the evaluation tools provide an audit trail and evidence of all the activities of ethical assessment and how they have been undertaken.

2.2.5 Governance. Governance in Information Technology (IT) provides a framework for ensuring the effective and efficient use of IT support to reach organizational strategies and objectives [55]. It involves providing an organizational regulatory framework that takes into consideration the laws, best practices, and needs of the various stakeholders. Ethical needs of stakeholders have become an important feature of software projects.

In software development, governance assists with evaluating, directing, and monitoring processes along the software life cycle to ensure that the various software artifacts requiring governance are attended to [55]. Considering that software development has stakeholders with varying needs, IT governance does pay attention to ethical challenges related to governance to ensure that the software development is conducted ethically and responsibly [56]. Several authors propose various governance practices such as public engagement frameworks [32], ethics committees, accountability contracts, and standardized reporting [38] and policies [40]. The governance machinery provides an extra layer of oversight on ethics considerations in software development. This ensures that ethics are part of organization-wide strategies related to digital transformation and also that stakeholders participating at the governance level influence the ethical development of software.

The application of these ethical practices in software development can strongly support the achievement of the ethical needs of stakeholders. The software process has outcomes and artifacts at every stage. For example, these can include ensuring that during the analysis phase, the requirements of the software may include ethics. Furthermore, the designs such as architectural designs, software classes design, etc., also reflect how the ethics will be implemented. Furthermore, this may include the evaluation of all artifacts to determine if ethics have been accounted for.

3 Implications and conclusions

The contribution of research and practice on ethics considerations in software development is important. Through the review of the literature, this article managed to identify and analyze elements of ethical practices applicable to software development. The practices are organized as awareness, enforcement, decision-making, evaluation, and governance. A combination of these practices is useful for ensuring a broader approach to meeting ethical needs of software stakeholders in the development of software. These practices apply to the entire software process to assist in the achievement of its various outcomes contributing to ethical software development.

These practices were drawn mainly from empirical studies which have tested their applicability and usefulness. Therefore, in practice, a combination of these practices may be adopted to inform expansive ethics considerations in software development by both practitioners and software development organizations. Moreover, researchers involved in the development of ethics frameworks may integrate these practices into the frameworks to support ethical software development. Therefore, the identified practices can be used to form a comprehensive framework for ethics-aware software engineering practice.

The integration of the ethical needs of stakeholders in software development can lead to software success. Therefore, practices that are not useful in assisting software stakeholders to achieve ethics have the potential to limit successes that can be achieved in software development. The software engineering practice should ensure that apart from providing support to achieve the technical aspects of software development, enough attention should be provided to practices that allow for the integration of ethics in the software process to ensure that all software development outcomes are in line with the ethical needs of stakeholders.

Although this study has managed to identify and synthesize ethical practices useful for supporting ethical ethics-aware software engineering practice, the identification and analysis of these aspects of ethical practices are confined to literature generated by our searches. Secondly, the findings of this study are limited to literature contents, not an empirical study. The findings of this study can inform future empirical studies to expand further knowledge on the combined application of these ethical practices in software development.

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